

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Pharmacy

Semester: VI

Subject Name: Pharmaceutical Biotechnology

Subject Code: BP605T

Max Marks: 75

Date: 30/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10X2) = 20

- A) Write advantages & disadvantages of solid state fermentation.
- B) Define terms- A) Transformation B) Transduction
- C) Write the functions of MHC.
- D) Define vector. Write the features of good vector.
- E) Differentiate between monoclonal antibodies and polyclonal antibodies.
- F) What is enzyme immobilization? Enlist the methods of enzyme immobilization.
- G) What is the role of primer in PCR?
- H) Enlist the enzymes used in gene cloning.
- I) Name four mutagenic agents.
- J) What is genetic engineering?

Q.2. Long Answers (Answer 2 out of 3) (10X2) = 20

- A) Describe in detail different biotransformation reactions with special reference to steroids.
- B) Explain different types of biosensor and applications of biosensors used in diagnostics.
- C) Explain in detail the general methods of recombinant DNA technology.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- A) Describe in brief structure of immunoglobulin.
- B) Write note on generic organization of eukaryotes.
- C) Explain the principle and applications of ELISA test.
- D) Explain in detail production of hepatitis B vaccine by recombinant technology.
- E) What is basis of type-II hypersensitivity reaction? Explain with suitable example.
- F) Explain the principle of protein engineering with neat labeled flow chart.
- G) Describe in brief different types of fermenters.
- H) What is PCR? Explain the applications of PCR.
- I) Write in brief production and importance of different plasma substitutes.

*** END OF THE PAPER ***